HVAC Building Energy Management via Multi-Objective Deep Reinforcement Learning

Complete Research Report: Phases 0–3
All Experiments, Problems, Solutions, and Breakthroughs

Almaz Sapargali
DigitAlem LLP, Almaty, Kazakhstan
Project IRN AP23488794

March 2026

Abstract

This report documents the complete research trajectory toward deploying a Multi-Objective Reinforcement Learning (MORL) system for HVAC building energy management, spanning four phases and over fifteen experiments. Phase 0 establishes a PPO baseline with vector rewards (28% energy reduction). Phase 1 develops a physics-informed RC Neural ODE surrogate achieving 32,313× measured speedup with RMSE = 0.163°C ($R^2 = 0.991$). Phase 2 solves an inverse problem for building calibration (79% RMSE improvement, C_{zon} recovery with 7.9% error). Phase 3 conducts eleven MORL experiments—including reward shaping, GRU weather forecasting (Gao et al.), Runtime Shielding (Xu et al.), SAC, HDRL (Liao et al.), real weather integration, Emergency Heating Mode, expanded action spaces, and BOPTEST fine-tuning—culminating in $m_s = 0.697 \pm 0.368$ across 12 monthly scenarios with summer $m_s = 0.140$–0.187.

A thermostatic ablation study (Gao et al. comparison) reveals that the compact 9-feature architecture (RMSE = 0.73°C in July) outperforms richer 17-feature variants on BOPTEST, confirming the sim-to-real gap as the bottleneck. The final breakthrough—direct T_{supply} control trained on a dedicated surrogate v3—reduces mean RMSE from 4.70°C to 0.84°C and winter RMSE from 8.89°C to 0.78°C (11× improvement), proving that the "structural gap" was caused by BOPTEST's PI controller, not building physics. In total, 15 technical problems are documented with root causes and solutions. All code, models, and data are reproducible.

Contents

1 Introduction and Motivation

Building HVAC systems account for approximately 40% of global primary energy consumption [1, 5]. Traditional controllers—PI, rule-based, and MPC—face fundamental limitations: PI controllers cannot optimize competing objectives, rule-based systems lack adaptability, and MPC requires accurate building models and is computationally expensive at deployment (2.67 s per step in [1]).

Deep Reinforcement Learning (DRL) offers a data-driven alternative that can learn complex control policies from interaction. However, three critical challenges remain:

1. **Training speed:** High-fidelity simulators like BOPTEST [9] run at ∼37 steps/second, requiring 73.9 hours for 10M training steps.
2. **Sim-to-real transfer:** Policies trained on surrogate models fail when deployed on the target environment.
3. **Seasonal robustness:** Most studies validate on a single scenario (e.g., peak heat week [1]), hiding poor performance in other seasons.

Our research program addresses all three challenges through four phases, documented below with every problem encountered, every solution attempted, and every result obtained.

2 Phase 0: MORL-PPO Stabilization

2.1 Objective

Establish a working multi-objective PPO agent on BOPTEST with vector rewards $\mathbf{r} = [r_{\text{comfort}}, r_{\text{energy}}]$ and scalarized objective $R = \omega_c r_c + \omega_e r_e$.

2.2 Problem P0.1: Scale Dominance

Root cause: Initial energy scaling $\alpha = 10^{-7}$ made r_{energy} approximately $10^5\times$ smaller than r_{comfort}. The agent ignored energy entirely.

Solution: Set $\alpha = 1/P_{\max} = 2 \times 10^{-4}$, equalizing the magnitude of both reward components.

2.3 Problem P0.2: Observation Space Mismatch

Root cause: Observation space defined as $\text{Box}[-10^6, 10^6]$ but actual observations normalized to $[-1, 1]$. PPO's value function could not learn meaningful estimates.

Solution: Observation space set to $\text{Box}[-1, 1]$.

2.4 Result

After fixes, a Pareto sweep across 5 weight configurations ($\omega_c \in \{0.2, 0.5, 0.8, 0.9, 1.0\}$) demonstrated 28% energy reduction (192.3 → 138.7 kWh) with a monotonic Pareto front. This validated the MORL formulation.

3 Phase 1: RC Neural ODE Surrogate

3.1 Objective

Replace the slow BOPTEST simulator (37.6 steps/s) with a fast, differentiable surrogate for DRL training. Inspired by [1]'s linear RC model but extended to a nonlinear Neural ODE.

3.2 Architecture

Physics-informed residual structure mirrors thermal inertia:

$$T(t+1) = T(t) + 10 \cdot f_\theta(T_{\text{zone}}, T_{\text{amb}}, h, d, a_0, a_1) \tag{1}$$

$$P_{\text{total}} = \text{Softplus}(g_\phi(\mathbf{x})) \cdot P_{\max} \tag{2}$$

The residual form in Eq. (1) constrains the network to predict ΔT (small corrections), not absolute temperature. The Softplus in Eq. (2) enforces $P \geq 0$. Total: 8,482 trainable parameters.

3.3 Problem P1.1: Thermal Capacitance Too Large

Root cause: Initial $C_{\text{zon}} = 10^7$ J/K was 20× too large, causing extremely slow temperature dynamics.

Solution: Data-driven estimate from BOPTEST step responses: $C_{\text{zon}} = 5.3 \times 10^5$ J/K.

3.4 Problem P1.2: Parameter Drift

Root cause: C_{zon} was a trainable parameter that drifted during optimization, destabilizing the model.

Solution: PyTorch `register_buffer` — C_{zon} participates in forward pass but is not updated by the optimizer.

3.5 Problem P1.3: Neural Network Cancels HVAC Signal

Root cause: The NN learned $f_\theta \approx -Q_{\text{HVAC}}$, effectively canceling the heating/cooling input. The model predicted constant temperature regardless of action.

Solution: Predict ΔT directly via the residual structure (Eq. (1)), removing the opportunity for the NN to "undo" the physics.

3.6 Speed Benchmark

Table 1: Measured speed comparison (100 BOPTEST steps, 100k surrogate steps)

Backend	Speed [steps/s]	10M steps	Speedup
BOPTEST (HTTP API)	37.6	73.9 hours	1×
Surrogate (single)	6,921	24.1 min	184×
Surrogate (batch=32)	107,470	93 sec	2,858×
Surrogate (batch=1024)	1,215,752	8.2 sec	**32,313**×

10M steps at 1 hour/step = 1,142 years of simulated building operation.

3.7 Result

Violation rate improved from 78.7% to 35.3%; m_s improved from 1.284 to 0.510.

4 Phase 2: Inverse Problem — Surrogate Calibration

4.1 Objective

Adapt the surrogate to a new building by solving an inverse problem: recover C_{zon} and R_{wall} from telemetry data contaminated with noise ($\sigma = 0.3$°C), latency ($\tau \in \{1, 2, 3\}$ steps), and bias (+0.5°C).

4.2 Problem P2.1: Latency Collapses Signal

Root cause: Sensor latency reduced the variance of ΔT from 9.48 to 0.42°C, making the optimization ill-posed — all parameter values produced similar loss.

Solution: Preprocess each telemetry artifact (noise, latency, bias) separately before joint calibration.

4.3 Problem P2.2: Linear Calibration Insufficient

Root cause: Adjusting only C_{zon} and R_{wall} improved RMSE by only 5.2% — the nonlinear dynamics captured by the NN weights were more important.

Solution: Fine-tune all NN weights jointly with dual learning rates: $\eta_{\text{NN}} = 10^{-4}$ for network weights, $\eta_{\text{calib}} = 10^{-2}$ for physical parameters.

4.4 Result

RMSE: 9.12 → 1.91°C (79% improvement). C_{zon} recovered with 7.9% error.

5 Phase 3: Safe MORL — Surrogate v2 and Training

5.1 Surrogate v2 Development

Four problems were resolved to create a production-quality surrogate:

P3.1 (Missing T_{amb}): The original surrogate had no ambient temperature input. Solution: 8-input architecture including T_{amb}, hour, and day.

P3.2 (Single-step loss): Training on single-step predictions allowed error accumulation during multi-step rollouts. Solution: multi-step loss with $K = \{2, 4\}$.

P3.3 (Insufficient data): Original 10,000 samples covered limited operating conditions. Solution: 51,200 samples from 4 policies × 4 seasons on BOPTEST.

P3.5 (Wrong BOPTEST key): Ambient temperature was read from the wrong sensor key. Solution: payload inspection revealed correct key `zon_weaSta_reaWeaTDryBul_y`.

Result: RMSE: 1.128 → 0.163°C (6.9× improvement), $R^2 = 0.991$.

5.2 Safety Filter

Horizon-2 predictive filter with 0.82°C margin. Gradient MPC fallback ($\lambda_S = 50$, 50 Adam iterations).

P3.4: Horizon 4 was too conservative (93% action rejection). Solution: horizon 2 with wider margin.

6 Phase 3: Eleven MORL Experiments

6.1 Experiment 1: PPO Baseline (Synthetic Weather)

Setup: 10M steps, 32 parallel surrogate envs, synthetic sinusoidal T_{amb} with domain randomization ($T_{\text{amb,base}} \sim U[-10, 25]$°C).

Result: Mean $m_s = 0.798 \pm 0.292$, violation 51.9% (10 scenarios).

6.2 Experiment 2: Reward Shaping (P3.7 — Failed)

Idea: Asymmetric penalties (cold ×1.5), quadratic for deviations > 2°C, free energy during violations.

Result: Violation *increased* to 81.8%. The shaped reward amplified the sim-to-real gap by training the agent to exploit surrogate-specific dynamics that don't exist on BOPTEST.

Lesson: Reward shaping is dangerous when there is a sim-to-real gap.

6.3 Experiment 3: GRU Weather Forecast — Gao et al. (P3.8 — Partial)

Idea from [3]: Augment observation with 24-hour weather forecast processed by GRU(32). Architecture: state(4) → MLP(64), forecast(24) → GRU(32), concat → MLP(64,64).

Result: July improved −8.3 pp, but January worsened +4.5 pp. Mean $m_s = 0.893$.

Lesson: Surrogate-generated forecasts ≠ real weather patterns.

6.4 Experiment 4: Runtime Shielding — Xu et al. (P3.9 — Failed)

Idea from [4]: At deployment, iteratively correct unsafe actions ($\Delta a = 0.1$, up to 20 iterations) using surrogate predictions.

Result: Best at margin $\delta = 0$: $m_s = 0.878$, but energy doubled.

Lesson: Shield corrections are based on surrogate predictions, which are inaccurate on BOPTEST.

6.5 Experiment 5: SAC Algorithm (P3.11 — Failed)

Idea: Soft Actor-Critic (off-policy) might achieve better sample efficiency than PPO (on-policy).

Result: 3M steps, 8 envs. Mean $m_s = 1.063$, violation 74.3%. SAC overheated all seasons ($T_{\max}$ up to 29.9°C).

Lesson: PPO outperforms SAC for this HVAC control task.

6.6 Experiment 6: HDRL with Synthetic Weather — Liao et al. (P3.12)

Idea from [6]: Two specialized PPO agents with threshold-based meta-controller:

- Winter agent trained on $T_{\text{amb,base}} \sim U[-15, 5]$°C (5M steps, 16 envs)
- Summer agent trained on $T_{\text{amb,base}} \sim U[10, 30]$°C (5M steps, 16 envs)
- Meta-controller: $T_{\text{amb}} < 12$°C → winter, else summer

Result: Aug $m_s = 0.223$ (best individual at that point), mean $m_s = 0.834$.

Lesson: Seasonal specialization significantly improves summer performance.

6.7 Experiment 7: Real Weather Integration

Key innovation: Replaced synthetic sinusoidal T_{amb} with real Belgian weather data from 51,200 BOPTEST measurements, organized into a 366×24 hourly lookup table.

Results:

- PPO + Real Weather: violation 48.8%, $m_s = 1.000$
- HDRL + Real Weather: violation 45.6%, three scenarios below 5%, but $m_s = 0.972$ due to extreme cold events ($T_{\min} = -4.5$°C in January)

Lesson: Real weather reduces the weather mismatch component of the sim-to-real gap, but exposes the extreme event problem.

6.8 Experiment 8: Emergency Heating Mode

Problem: Extreme cold fronts caused T_{zone} to crash to -4.5°C, driving r_{sev} to 1.21.

Solution: Rule-based override at inference time (no retraining):

$$a_t = \begin{cases} [1.0, 1.0] & \text{if } T_{\text{amb}} < 5°\text{C} \wedge T_{\text{zone}} < 20°\text{C} \quad \text{(Emergency)} \\ \pi_{\text{winter}}(s_t) & \text{if } T_{\text{amb}} < 12°\text{C} \\ \pi_{\text{summer}}(s_t) & \text{otherwise} \end{cases} \tag{3}$$

Result: $T_{\min}$ raised from -4.5°C to 13.2°C. Winter r_{sev} reduced by 3.3×. Mean $m_s = 0.697 \pm 0.368$.

Lesson: Combining learned policies with rule-based safety overrides outperforms either approach alone.

6.9 Experiment 9: 5-Feature Obs + Expanded Action Space [18,30]°C

Idea: Add T_{amb} as 5th observation feature so agent "sees" outdoor temperature. Expand action mapping to $T_{\text{set}} \in [18, 30]$°C for pre-heating capability.

Result: $m_s = 0.704$, Sep $m_s = 0.140$ (best individual scenario). No significant improvement over 4-feature + [21,25]°C configuration because Emergency Mode already handles extreme events.

6.10 Experiment 10: BOPTEST Fine-tuning (Two Rounds)

Idea: Fine-tune HDRL agents directly on BOPTEST to calibrate action-response mapping.

Round 1: 50k steps, lr $= 10^{-4}$, 4-feature obs, [21,25]°C. `clip_fraction` = 0.000.

Round 2: 50k steps, lr $= 3 \times 10^{-4}$, 5-feature obs, [18,30]°C. `clip_fraction` = 0.000.

Conclusion: Policy completely frozen in both rounds. The remaining gap was initially attributed to "structural action-response mismatch" — the building physics allegedly could not maintain comfort. This conclusion was later overturned by Experiment 11.

6.11 Experiment 11: Direct $\mathrm{T}_{\text{supply}}$ Control

Idea from [2, 4]: Instead of controlling T_{set} (interpreted by BOPTEST's PI controller), control the supply air temperature T_{supply} directly.

Implementation: Required a complete retraining cycle:

1. Collected 51,200 BOPTEST steps with $T_{\text{supply}} \in [18, 35]$°C control (31.2 min)
2. Trained surrogate v3 on $\mathrm{T}_{\text{supply}}$ data (RMSE = 0.626°C, $R^2 = 0.979$)
3. Trained thermostatic PPO on v3 surrogate (17-feature obs, 49.1 min)
4. Validated on BOPTEST with direct `fcu_oveTSup_u` control

Critical BOPTEST configuration: Set $T_{\text{set,cool}} = 40$°C and $T_{\text{set,heat}} = 15$°C to prevent the PI controller from interfering with direct $\mathrm{T}_{\text{supply}}$ commands.

6.12 Best MORL Configuration (Experiments 1–10)

6.13 All MORL Experiments Ranked

7 Thermostatic Ablation Study (Gao et al. Comparison)

To validate our Neural ODE surrogate against the thermostatic benchmark of [3], we trained single-objective agents to track $T_{\text{target}} = 22$°C with Gaussian reward (no energy penalty). Three variants were tested, progressively increasing observation complexity.

Table 2: Best MORL result: HDRL + Real Weather + Emergency Mode (12 months)

Scenario	Viol.%	$T_{\min}$	$T_{\max}$	E [kWh]	m_s	W%	E%
Jan_Winter	73.2	13.2	25.0	90.2	1.102	33	61
Feb_Winter	74.7	13.4	25.3	79.3	1.110	33	62
Mar_Spring	72.0	15.3	25.3	80.4	0.990	35	59
Apr_Spring	63.7	15.1	24.8	56.0	0.919	41	41
May_Spring	46.4	16.7	24.6	37.9	0.669	54	10
Jun_Summer	19.0	16.5	25.1	68.7	0.405	22	2
Jul_Summer	6.0	19.0	28.2	107.9	0.187	1	0
Aug_Summer	5.7	18.8	27.5	127.0	0.160	0	0
Sep_Autumn	5.7	18.6	28.0	172.6	0.178	6	0
Oct_Autumn	36.3	16.1	26.5	124.6	0.594	41	16
Nov_Autumn	66.7	15.9	25.8	88.5	0.911	39	50
Dec_Winter	80.7	14.1	25.2	76.0	1.137	29	68
Mean ± std	**45.8**			**92.4**	$\mathbf{0.697 \pm 0.368}$		

Table 3: All eleven MORL experiments

Method	m_s	Viol.%	Best	Key Finding
HDRL+RW+Emergency	**0.697**	45.8	Aug 0.160	Best mean
HDRL+RW+E+5F+[18,30]	0.704	46.8	Sep 0.140	Best scenario
PPO baseline (synth.)	0.798	51.9	Jul 0.357	Baseline
HDRL (synthetic)	0.834	60.3	Aug 0.223	Specialization
Shield (Xu)	0.878	64.5	Aug 0.254	Needs accuracy
GRU (Gao)	0.893	57.4	Jul 0.272	Forecast gap
HDRL + Real Weather	0.972	45.6	Aug 0.146	r_{sev} issue
PPO + Real Weather	1.000	48.8	Jul 0.237	Weather helps
Reward Shaping	1.050	81.8	—	Amplifies gap
SAC	1.063	74.3	Jun 0.828	PPO > SAC

7.1 Variant 1: Compact (9 features)

Obs: $[T_{\text{zone}}, \text{CO}_2, P_{\text{cool}}, P_{\text{fan}}, T_{\text{amb}}, h_{\text{norm}}, d_{\text{norm}}, T^{+1\text{h}}_{\text{amb}}, T^{+3\text{h}}_{\text{amb}}]$. Gaussian reward with smoothness penalty. Trained 10M steps in 44.4 min.

7.2 Variant 2: Extended forecast (12 features)

Added +6h, +12h, +24h forecast horizons to match Gao's 24h GRU. Asymmetric cold penalty (1.5×).

7.3 Variant 3: Rich temporal (17 features)

Added cyclic time encoding (sin/cos), previous action, ΔT_{zone} history. Non-saturating reward with deadband bonus, extreme penalty ($T < 19$°C), micro energy penalty. Network 256×256.

7.4 Variant 4: Direct $\mathbf{T_{supply}}$ (17 features, v3 surrogate)

Same 17-feature architecture but trained on surrogate v3 (T_{supply} data). At BOPTEST deployment, controls `fcu_oveTSup_u` directly instead of T_{set}.

7.5 Results Comparison

Table 4: Thermostatic control: all variants on BOPTEST

	v1 (9F)	v2 (12F)	v3 (17F)	v4 (T_{supply})
Control	T_{set}	T_{set}	T_{set}	T_{supply}
Surrogate	v2	v2	v2	v3
Jul RMSE	**0.73**	0.81	0.90	1.00
Aug RMSE	—	—	0.81	0.81
Winter RMSE	8.89	8.88	9.04	**0.78**
Mean RMSE	4.70	4.71	4.26	**0.84**
±1°C (all)	41%	40%	36%	**73%**
Best month	Jul 0.73	Jul 0.81	Aug 0.81	**Apr 0.63**
Training	44 min	40 min	49 min	49 min

7.6 The Breakthrough: v4 Direct T_{supply}

Table 5: v4 Direct T_{supply}: complete 12-month BOPTEST validation

Scenario	RMSE	±1°C	±0.5°C	E [kWh]
Jan_Winter	0.83	77%	50%	412
Feb_Winter	0.66	88%	51%	405
Mar_Spring	0.78	80%	46%	314
Apr_Spring	**0.63**	87%	60%	233
May_Spring	0.82	74%	28%	150
Jun_Summer	0.99	54%	24%	133
Jul_Summer	1.00	57%	29%	139
Aug_Summer	1.06	58%	25%	153
Sep_Autumn	0.97	69%	28%	187
Oct_Autumn	0.91	71%	24%	202
Nov_Autumn	0.63	88%	59%	274
Dec_Winter	0.84	74%	45%	431
Mean (all)	**0.84**	**73%**	39%	253
Winter	**0.78**	79%	49%	416
Summer	0.99	60%	27%	142

Key finding: The "structural gap" that plagued all previous experiments was **not** a building physics limitation. It was caused by **BOPTEST's PI controller**, which:

- Interpreted T_{set} conservatively, limiting heating power
- Responded slowly to setpoint changes, causing temperature lag
- Could not deliver maximum power even when $T_{set} = 25$°C (its maximum in our mapping)

Bypassing the PI controller with direct T_{supply} control reduced winter RMSE by 11× (8.89 → 0.78°C), proving the building has sufficient heating capacity when controlled directly.

Table 6: Final comparison with [3]

	Gao M.3 (PPO+GRU)	**Ours v1 ($T_{\mathbf{set}}$)**	**Ours v4 ($T_{\mathbf{supply}}$)**
Best RMSE	≈ 0.5°C	0.73°C	**0.63°C**
Mean RMSE	≈ 0.5°C	4.70°C	**0.84°C**
Winter RMSE	not reported	8.89°C	**0.78°C**
Architecture	GRU(32)	MLP (default)	MLP (256×256)
Obs features	28 (GRU)	9 (lookup)	17 (cyclic+forecast)
Training time	∼hours	44 min	49 min
Speedup	1×	32,313×	32,313×
Control level	T_{set}	T_{set}	T_{supply} (direct)

Table 7: Safety metric comparison with [1]

Controller	m_s	**Viol.%**	**Validation**
PI Controller	0.096	9.3%	Peak heat week
MPC	0.016	1.2%	Peak heat week
Safe DRL (DDQN+MPC)	0.000	0.0%	Peak heat week
Our MORL (yearly)	0.697	45.8%	12 months
Our MORL (summer)	0.160–0.187	5.7–6.0%	Jul/Aug/Sep
Our v4 thermostatic	≈ 0.05*	≈ 5%*	12 months

*Estimated from RMSE = 0.84°C with [21,25]°C bounds — formal m_s computation pending.

7.7 Comparison with Gao et al.

8 Comparison with Wang et al.

9 Root Cause Analysis: Three Components of the Sim-to-Real Gap

Our systematic experimentation decomposed the sim-to-real gap into three independent components:

1. **Weather mismatch (SOLVED):** Synthetic sinusoidal T_{amb} did not capture cold fronts, heat waves, or seasonal asymmetries. *Solution:* Real Belgian weather lookup table from 51,200 BOPTEST measurements. *Impact:* Violation reduced from 51.9% to 45.6%.

2. **Extreme event response (SOLVED):** The agent could not prevent deep temperature crashes during sudden cold fronts. *Solution:* Rule-based Emergency Heating Mode ($T_{\text{amb}} < 5$°C $\wedge$ $T_{\text{zone}} < 20$°C $\rightarrow$ max heat). *Impact:* $T_{\min}$: $-4.5 \rightarrow 13.2$°C; m_s: $0.972 \rightarrow 0.697$.

3. **PI controller bottleneck (SOLVED in v4):** Initially misidentified as "structural action-response mismatch." BOPTEST fine-tuning showed `clip_fraction` = 0, suggesting the gap was unfixable. However, Experiment 11 proved that bypassing the PI controller with direct T_{supply} control reduced winter RMSE from 8.89°C to 0.78°C (11×). The building has sufficient heating capacity; the PI controller was the bottleneck.

Methodological lesson: A "structural" limitation diagnosed through indirect evidence (frozen policy gradients) can be misleading. Direct experimental evidence (changing the control interface) is necessary to distinguish between true physical limitations and software/controller limitations.

10 Complete Problem Log (15 Problems)

Table 8: All 15 problems with root causes and solutions

ID	Root Cause	Solution	Impact
P0.1	Energy scale 10^{-7}	$\alpha = 1/P_{\max}$	Pareto front restored
P0.2	Obs Box$[-10^6]$	Box$[-1, 1]$	Value function stable
P1.1	$C_{\text{zon}} = 10^7$	Data estimate 5.3×10^5	Physics correct
P1.2	C_{zon} drifts	`register_buffer`	Parameter stable
P1.3	NN cancels HVAC	Residual ΔT	Architecture fix
P2.1	Latency ill-poses	Separate preprocess	RMSE 79% better
P2.2	Linear calib 5.2%	Joint NN fine-tune	C_{zon} 7.9% error
P3.1	No T_{amb} input	8-input architecture	Weather-aware
P3.2	1-step loss	Multi-step K=$\{2, 4\}$	Rollout stable
P3.3	10k data samples	51k (4×4)	Full coverage
P3.4	Horizon 4 strict	Horizon 2	2.2% false-safe
P3.5	Wrong BOPTEST key	Payload inspection	T_{amb} correct
P3.7	Reward amplifies gap	Reverted to original	Insight
P3.8	Forecast $\neq$ weather	Real weather lookup	Solved
P3.12	Flat $\neq$ seasons	HDRL + Emergency	m_s -46%

11 Progress Summary

Table 9: Complete improvement trajectory

Stage	m_s / RMSE	Δ	Key Innovation
Phase 0 initial	$m_s = 1.284$	—	MORL formulation
Phase 1 surrogate	$m_s = 0.510$	-60%	Neural ODE (32,313×)
PPO baseline	$m_s = 0.798$	-38%	10M steps, 32 envs
HDRL + RW + Emergency	$m_s = 0.697$	-46%	Seasonal + rule-based
Summer best (Sep)	$m_s = 0.140$	-89%	Summer specialization
Thermostatic v1	RMSE = 0.73°C	—	Gao comparison (summer)
T_{supply} v4	**RMSE = 0.84°C**	—	**PI controller bypass**
v4 winter	**RMSE = 0.78°C**	**11×**	**Structural gap solved**

12 Proposed Next Steps

1. **MORL with direct T_{supply}:** Combine the HDRL + Emergency architecture with T_{supply} control. Expected: $m_s < 0.3$ across all seasons.

2. **PMV/PPD comfort metric:** Replace temperature-only metric with ASHRAE Standard 55 [2], incorporating humidity, air velocity, and metabolic rate.

3. **PPG algorithm:** Phasic Policy Gradient [7] for improved fine-tuning convergence (separated policy/value updates).

4. **Optuna hyperparameter sweep:** Exploit 32,313× speedup for exhaustive search over learning rate, network architecture, reward weights.

5. **Edge deployment:** NVIDIA Jetson Orin Nano with real building IoT sensors for Phase 5.

13 Conclusion

This work established a complete MORL-based HVAC control pipeline through four phases and over fifteen experiments:

- **Physics-informed surrogate:** RC Neural ODE with 32,313× speedup (RMSE = 0.163°C, R^2 = 0.991), enabling rapid experimentation (35 min for 10M steps vs 73.9 hours on BOPTEST).

- **Inverse problem:** Gradient-based calibration recovering C_{zon} with 7.9% error from noisy, latent, biased telemetry.

- **MORL architecture:** Three-layer controller (Emergency → HDRL → Seasonal agents) achieving $m_s = 0.697$ across 12 months, with summer $m_s = 0.14$–0.19.

- **Sim-to-real gap decomposition:** Three components identified—weather mismatch (solved), extreme events (solved), and PI controller bottleneck (solved via direct T_{supply}).

- **Critical finding:** The "structural gap" diagnosed through frozen policy gradients (`clip_fraction` = 0) was not a building physics limitation but a **control interface limitation**. Direct T_{supply} control reduced winter RMSE from 8.89°C to 0.78°C (11× improvement), proving the building has sufficient capacity when the PI controller is bypassed.

The overall trajectory—from $m_s = 1.284$ (Phase 0) to $m_s = 0.697$ (MORL, −46%) to RMSE = 0.84°C (thermostatic, all seasons)—demonstrates that combining physics-informed surrogates, hierarchical architectures, and systematic gap analysis produces deployable building controllers.

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